

React.js Test Document

This document is intended for testing Retrieval-Augmented Generation (RAG) applications built with LangChain and ChromaDB.

What is React?

React is an open-source JavaScript library developed by Meta for building user interfaces. It follows a component-based architecture and is commonly used to build single-page applications.

Key Features

React provides reusable components, a virtual DOM for efficient rendering, JSX syntax, one-way data flow, hooks, and a rich ecosystem.

React Hooks

Common hooks include `useState`, `useEffect`, `useMemo`, `useCallback`, `useRef`, `useContext`, and `useReducer`.

Component Types

Modern React applications primarily use functional components. Class components are still supported but are less common in new projects.

Virtual DOM

The virtual DOM is a lightweight in-memory representation of the real DOM. React compares virtual DOM trees and updates only the necessary parts of the real DOM.

`useEffect`

The `useEffect` hook performs side effects such as API requests, event listeners, timers, and subscriptions after a component renders.

State Management

React applications can manage state using `useState`, Context API, Redux, Zustand, Jotai, or TanStack Query for server state.

Routing

React Router is the most popular library for client-side routing in React applications.

Performance Tips

Use `React.memo`, `useMemo`, `useCallback`, lazy loading, code splitting, and list virtualization for better performance.

Interview Questions

Explain the Virtual DOM. Explain `useEffect` dependency array. Difference between state and props. Controlled vs uncontrolled components. What are hooks?